



■ **Security**

Indian Railways hack: 3 Crore passengers' data leaked and is up for sale on the dark web

Data of over 3 Crore Indian Railways users has been leaked and is reportedly up for sale on the dark web. The data, which includes booking data as well as user information, is being sold for \$400 per copy, with buyers only allowed to buy five copies. In case a buyer wants exclusive access to the data, they will have to pay \$1,500 and \$2,000. The breach took place on December 27, 2022. On being questioned about the leak, the IRCTC said in a statement, "An incident regarding Indian Railway data breach has been reported in the media. In this connection it may be submitted that Railway Board had shared a possible data breach incident alert of CERT-In to IRCTC reporting a data breach pertaining to Indian Railways passengers. On analysis of sample data it is found that the sample data key pattern does not match with IRCTC history API. Reported/suspected data breach is not from the IRCTC servers. Further Investigation on the data breach is being done by IRCTC. All IRCTC Business Partners have been asked to immediately examine whether there is any data leakage from their end and apprise the results along with corrective measures taken to IRCTC." The name of the hacker is 'Shadow hacker', as per the posts made on forums where the data is up for sale. Nothing else is known about the entity.

■ **R&D**

Apple Watch ready for stress prediction: Here's all you need to know

With every release of the Apple Watch, the Cupertino-based tech giant Apple has been trying to pack in more and more health-focused features into their smartwatches. They have been a fan favourite and have prompted more and more individuals to take the plunge into the world of smartwatches. Now, a study has shown that the sensor data obtained from the Watch could be used to develop a stress prediction tool. Ever since the launch of the Apple Watch in 2014, there have been incremental updates added to both the software and hardware capabilities to ensure that there is an increase in the suite of health features that the watch has to offer. The latest Apple Watch Series 8 and Apple Watch Ultra come with Blood Oxygen, ECG App, and Heart Rate monitoring. Additionally, there are tracking features like Irregular rhythm notification, temperature sensing, and cycle tracking with retrospective ovulation estimates.



Coming back to the study that highlighted the possibilities that could be explored using the data from the Apple Watch, it was done by researchers from the University of Waterloo, Canada and was titled – Using apple watch ECG data for heart rate variability monitoring and stress prediction: A pilot study. There, the researchers worked with data collected from 33 study participants, which was modelled using the Random Forest (RF) and Support Vector Machines (SVM) techniques.

Commenting on the finding, the researchers wrote, "Our models showed specificity in their capacity to assess "no stress" states but were less successful at capturing "stress" states. Overall, the results presented here suggest that, with further development and refinement, Apple Watch ECG sensor data could be used to develop a stress prediction tool." There are multiple tracking apps available on the Apple App Store for users to look into and get started on their own as well! While these findings look to be promising, there are still a lot of possibilities left to be explored. As pointed out in the study as well, data collected from further research into this could also help authorities globally to form new and effective health policies too. **1**